Hard to Beat: Historical Rules versus Conditional Quantile Models for Peak Net-Position Limits

Abstract

Dealers monitor the largest absolute position reached before a review horizon ends. We compare conditional quantile models with historical rules using hourly FX positions, weekly New York Fed positions and dealer aggregates in 17 CFTC futures markets. Histories and update schedules are matched. Historical rules are hard to beat. First, FX regression lowers matched loss by 14.8% and 16.6% across two periods, but its ranking against a shorter rolling rule changes. Second, historical rules beat regression in 34 of 48 New York Fed comparisons and 29 of 34 CFTC comparisons; finite-penalty hindsight still favors history in most short weekly comparisons. They usually remain in the model confidence set alongside alternatives. In calibrated simulations, expanding the penalty menu cuts the constant-parameter median excess loss from 116.2% to near zero; switching adds no observed median loss in this short-history comparison. Third, in CFTC, lower scores need not mean lower limit costs: higher reserve charges and lower fixed breach charges generally favor expanded regression. The evidence is retrospective, and the original FX evaluation period had been inspected beforehand.

Keywords: Peak net positions; Quantile forecasting; Historical benchmarks; Forecast comparison; Calibration; Dealer positions.

1 Introduction

Dealer position limits concern the largest exposure reached before a review horizon ends, since an endpoint within the limit can conceal an earlier breach (Figure 1). We therefore compare historical and conditional forecasts of the sampled absolute-position peak, using the same target for every method.

Historical headroom combines the current magnitude with completed past peak behavior, while conditional models estimate additional relationships from finite historical windows. Matching histories and update schedules isolates the forecast gain associated with the added conditioning within each comparison.

Our evidence covers hourly FX positions, their daily aggregates, New York Fed categories and CFTC dealer-category futures positions. The two official weekly reporting systems describe different aggregate exposures, so their comparisons retain separate reference procedures and source-specific scales.

Three questions organize the evaluation.

RQ1 asks whether conditional quantile methods improve sampled peak forecasts with matched histories and update schedules.

RQ2 asks whether those gains extend across frequencies and independently reported dealer-position aggregates.

RQ3 asks whether score gains imply lower limit costs under reserve, breach and exceedance preferences.

The paper makes three empirical findings.

First, matched FX gains recur, but rankings against a shorter rolling rule change. The period comparisons separate conditioning gains from the reference procedure (Table 4). Delayed calibration preserves the later matched gain while changing breach behavior. Within the original

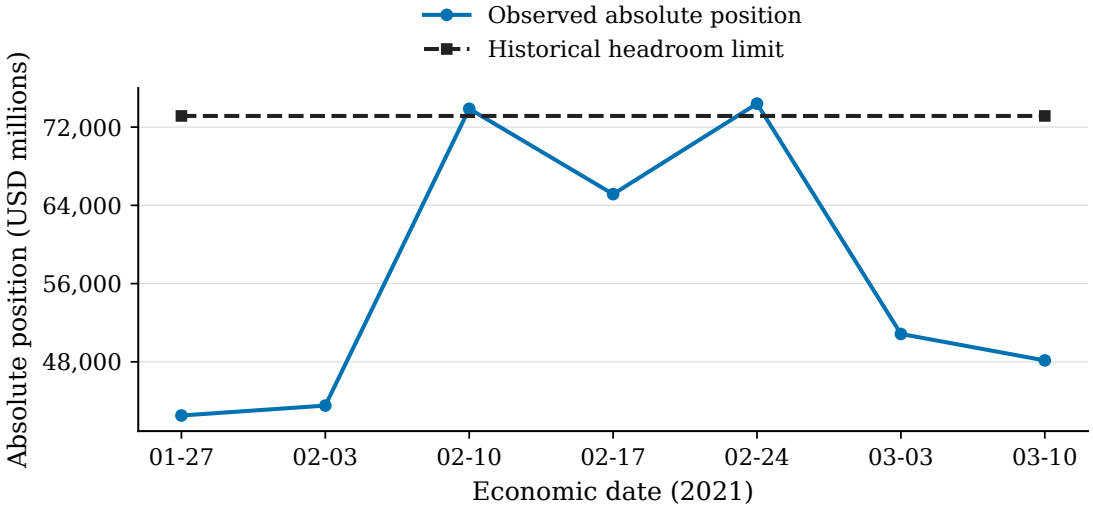


Figure 1: A selected Treasury-bill path that returns within the reported historical limit at its endpoint after an intervening breach. The rule chooses the earliest eligible original evaluation window. Endpoint containment can conceal a peak breach.

FX period, the full-regression learning curve improves at intermediate histories but gives back part of the gain at the longest history (Figure 2).

Second, historical rules remain competitive in weekly data, where both penalty selection and the finite menu leave losses. The panel and independent CFTC reporting system favor historical controls in most comparisons, and historical methods usually survive joint forecast comparison (Tables 5–7). Even the best finite penalty chosen with evaluation hindsight loses in 29 of 41 short-history weekly comparisons (Section 7.4). Longer validation narrows CFTC losses and selects the intercept-only limit more often, linking tuning instability to the available validation history. Expanded simulation menus also bring short-history regression near its historical reference, with no additional observed median loss from parameter switching. These retrospective diagnostics distinguish a finite-menu gap from selection errors, without ruling out other conditional signals.

Third, higher reserve charges and lower fixed breach charges generally favor expanded regression in CFTC. The direction has local exceptions (Figure 5). Lower scores therefore need not mean lower limit costs, and validation should report reserve amounts, breach counts and exceedance amounts together.

Section 3 defines the response. Section 7 answers RQ1 and RQ2; Section 8 answers RQ3. Elementary floor and endpoint properties support the forecast definitions. The forecasting, comparison and cost-accounting procedures use standard methods.

2 Related literature

Earlier FX studies forecast signed next-hour position intervals using historical states, calendar information and local prediction methods (Wisniewski et al., 2020; Wisniewski, 2024). Their chronological training, calibration and test design reports scores and coverage together, providing a natural comparison for the present forecasting task. Our peak response instead uses a complete future path segment and the current magnitude, producing different labels and limits.

Path-risk forecasts address within-period extremes. MaxVaR studies the worst cumulative portfolio return under fixed holdings and diffusion assumptions (Boudoukh et al., 2004). The

payment-liquidity thesis forecasts conditional maximum deficits from confidential transactions (van den Hoogen, 2017). Settlement studies examine tail liquidity needs and collateral requirements (Petrunia et al., 2019; Arjani et al., 2020). Duffie and colleagues study dealer capacity and Treasury-market functioning, including PnL quantile forecasts (Duffie et al., 2023). These responses differ from the sampled absolute-position peak.

Regression quantiles target asymmetric forecast errors directly (Koenker & Bassett, 1978), while autoregressive quantiles and quantile forests supply other conditioning structures (Engle & Manganelli, 2004; Meinshausen, 2006). Consistent scores and elementary-score representations permit comparison across evaluation preferences (Gneiting, 2011; Ehm et al., 2016). Table 1 locates the present peak response, completed labels and matched historical controls.

Table 1: Related studies: reported targets, data units and evaluation approaches.

Study	Response	Data unit	Evaluation
Wisniewski et al. (2020)	Next-hour signed-position interval	One market maker's hourly FX positions	Pinball and interval scores; coverage tests
Boudoukh et al. (2004)	Worst cumulative portfolio-return quantile	Fixed portfolio under a lognormal diffusion	MaxVaR/VaR ratios; discrete-monitoring simulation
van den Hoogen (2017)	Conditional quantile of cumulative payment-deficit peak	ABN AMRO transaction accounts	Coverage and independence backtests
Duffie et al. (2023)	Capacity, Treasury illiquidity and PnL tail quantiles	Dealer positions, transactions and PnL	Illiquidity regressions; out-of-sample PnL pseudo-R^2
Engle and Manganelli (2004)	Conditional return quantile	GM, IBM and S&P 500 daily returns	Quantile objective, exception rates and DQ tests
Ehm et al. (2016)	Quantile and expectile forecast comparison	Economic and wind-speed forecasts	Elementary-score curves and score differences
Meinshausen (2006)	Conditional response quantile	Regression datasets	Cross-validated quantile loss
This study	Sampled absolute-position peak	FX positions; weekly position aggregates	Pinball loss, breaches, model confidence sets and costs

The matched FX gain changes interpretation with its reference, while the weekly comparisons expose both tuning errors and finite-menu losses. Expanded simulation menus remove large short-history losses, with parameter switching adding no observed median loss in that comparison. Higher reserve charges and lower fixed breach charges generally favor regression in CFTC, although local exceptions remain. Paired tests support these forecast comparisons (Diebold & Mariano, 1995; Giacomini & White, 2006; Hansen et al., 2011), while inventory-sensitive quoting motivates exposure monitoring (Avellaneda & Stoikov, 2008). These findings concern passive reports on observed position paths, leaving their effect on subsequent trading decisions unmeasured.

3 The forecast target

The target is the largest sampled absolute position from issue time through the stated horizon, with N_t denoting signed position and $\mathcal{F}_t$ the issue-time information. The peak $Y_{t,H}$, observed floor B_t and nonnegative headroom $Z_{t,H}$ satisfy the following identities.

$$Y_{t,H} = \max_{0 \le h \le H} |N_{t+h}|, \qquad B_t = |N_t|, \qquad Z_{t,H} = Y_{t,H} - B_t \ge 0.$$

A feasible limit covers the current observed exposure, but Figure 1 shows that endpoint containment can still conceal an earlier breach. We score reports with pinball loss and count a strict breach when the peak exceeds the limit.

$$\rho_\tau(u) = \tau u_+ + (1-\tau)(-u)_+, \qquad I_{t,H} = \mathbf{1}\{Y_{t,H} > C_{t,H}\}.$$

Proposition 1 (known-floor projection). Suppose $Y \ge B$ almost surely and the issue information contains B. Projecting any forecast c to $\max\{B, c\}$ weakly improves pinball loss for every outcome, with the gain shown below. The true conditional lower inverse quantile at level $1-\alpha$ has strict exceedance probability at most α, including at atoms.

$$\rho_\tau(Y-c) - \rho_\tau(Y - \max\{B,c\}) = \tau(B-c)_+, \qquad \Pr(Y > q \mid \mathcal{F}_t) \le \alpha.$$

Proposition 2 (classical endpoint-to-peak bound). Let future real increments be independent and individually symmetric about zero, with partial sums S_h. Fix a starting position x, set $B = |x|$, and let q be the population $(1-\alpha/2)$ quantile of $|S_H|$. The triangle inequality and the classical Lévy maximal inequality imply the bound below. A conditional application requires these assumptions and the endpoint quantile under the law given issue-time information.

$$\Pr\left(\max_{0 \le h \le H} |x + S_h| > B + q\right) \le \Pr\left(\max_h |S_h| > q\right) \le 2\Pr(|S_H| > q) \le \alpha.$$

Supplement Sections S1–S2 retain the proofs and empirical qualifications, while Sections S3–S5 distinguish mean-reverting levels from serially correlated increments.

Separately fitted horizons can violate peak-quantile ordering, so Supplement Section S36 retains the original crossing diagnostic alongside the floor and nesting checks.

4 Data

The FX application uses the hourly Net Position Time Series Dataset, reporting market-maker client-flow positions (Lindsay, 2020; Wisniewski et al., 2020). Its 18,576 records retain source USD units; the later, earlier and full-prefix comparisons use distinct chronological issue grids with the same sampled-peak definition.

The weekly data are New York Fed Primary Dealer Statistics (Federal Reserve Bank of New York, n.d.). Reporting definitions follow the government-securities dealer framework and position-accounting instructions (Board of Governors of the Federal Reserve System, 2013, 2024). The Treasury-bill series has 706 economic-date observations in USD millions, measuring fair-value positions aggregated as long-minus-short totals. Additional categories pass metadata and calendar checks and use complete weekly grids, with separate fits by category.

Daily aggregation takes the last recorded position on each nonempty source-naive calendar day. It retains partial boundary days, yielding 930 days and 501 common evaluation issues, with the completed day's end serving as the issue clock. The resulting target is a peak over subsequent daily records.

The CFTC application retrospectively evaluates forecasts for a survivor universe using downloaded economic dates. It uses futures-only Dealer/Intermediary long positions minus short

positions (Commodity Futures Trading Commission, n.d.). The fixed screen requires uninterrupted report weeks from 2010, stable literal codes, exchange and units. It excludes documented consolidation-code families before ranking eligible contracts by full-sample mean open interest. Seventeen contracts pass the screen, below the planned maximum of twenty. Supplement Section S30 lists every selected code and explains the eligibility rules; complete code-level exclusions remain linked there.

There are 877 weekly positions through 20 October 2026. Validation uses 52 origins; evaluation uses the last 260 eligible origins with horizon six. The evaluation position dates run from 21 September 2021 to 8 September 2026, with the final peak label completing on 20 October 2026. Refits occur every thirteen records with histories of 52 and 260, using downloaded economic dates; first-release vintages remain unobserved. Positions usually refer to Tuesday and are released Friday, so the recorded position clock differs from the public information clock.

Table 2 describes the aligned source records and complete six-record peaks, with monetary series expressed in USD millions. Panel summaries take nine separate medians of within-series statistics, preserving the heterogeneous exposure units across categories and contracts. Within each forecasting comparison, every method uses identical peak labels, current floors and issue origins. The same-record FX aggregation holds dealer identity fixed while changing its measurement clock and sampled response.

Table 2: Data and matched six-record peak summaries.

Panel A. Sources, aligned samples and existing split dates

Data	Frequency	Series / rows	Unit	Sample dates	Validation	Evaluation
					2015-10-08	2016-01-01
		1 series			–2015-12-30	–2017-01-23
		18,576 rows		2014-02-05	2015-04-21	2015-07-15
FX, hourly	Hourly	18,570 peaks	USD mn	–2017-01-24	–2015-07-13	–2015-10-06
		1 series				
		930 rows		2014-02-05	2015-02-24	2015-05-24
FX, daily	Day end	924 peaks	USD mn	–2017-01-24	–2015-04-24	–2016-12-27
		1 series				
		706 rows		2013-04-03	2019-07-24	2021-01-06
Treasury bills	Weekly	700 peaks	USD mn	–2026-10-07	–2020-07-15	–2026-04-22
		12 series				
		706 rows		2013-04-03	2019-07-24	2021-01-06
NY Fed panel	Weekly	700 peaks	USD mn	–2026-10-07	–2020-07-15	–2026-04-22
		17 series				
		877 rows		2010-01-05	2020-08-11	2021-09-21
CFTC panel	Weekly	871 peaks	Contracts	–2026-10-20	–2021-08-03	–2026-09-08

Panel B. Within-series mean, sample SD and empirical 95th percentile

	$\lvert N\rvert$			Peak Y			Headroom Z		
Data	Mean	SD	95th	Mean	SD	95th	Mean	SD	95th
FX, hourly	8.92	10.2	27.6	11.1	11.7	30.2	2.15	5.92	7.96
FX, daily	9	10.5	27.5	17.9	16.6	41	8.91	14.8	24.9
Treasury bills	3.95×10^4	2.72×10^4	9.26×10^4	5.53×10^4	3.04×10^4	1.08×10^5	1.58×10^4	1.6×10^4	5×10^4
NY Fed panel	1.24×10^4	8.34×10^3	3.33×10^4	1.6×10^4	9.04×10^3	3.74×10^4	3.75×10^3	3.56×10^3	1.05×10^4
CFTC panel	6.02×10^4	4.6×10^4	1.48×10^5	8.18×10^4	4.56×10^4	1.69×10^5	1.99×10^4	2.09×10^4	6.06×10^4

Notes. All summaries use complete $H = 6$ origins and include the issue anchor; SD uses $m-1$, and the empirical 95th percentile has rank $\lceil 95m/100 \rceil$. Daily FX uses the last record of each nonempty source-naive calendar day at day end; the reported FX comparison periods were already inspected. Dollar values are in millions; panel entries are nine separate medians of within-series summaries, without exposure pooling or independence inference, and CFTC retains heterogeneous native contract counts.

5 Forecasting methods

Unless otherwise stated, forecasts target the conditional quantile at $\tau = 19/20$. The horizons $H \in \{6, 24\}$ count subsequent source records, and each peak includes the issue position. Calendar

gaps extend an FX horizon's elapsed span, while weekly and daily steps represent different review intervals under the same nineteen-to-one error-weight preference.

Historical headroom adds a corrected empirical quantile of completed headrooms to the observed magnitude:

$$F_{t,H} = B_t + Q_j\{Z_{i,H} : i \in \mathcal{T}_r\}, \qquad j = \lceil (m+1)\tau \rceil, \quad m = |\mathcal{T}_r|.$$

Here Q_j is the jth order statistic, and the training set contains only labels completed by refit time r. At the sensitivity level where this rank exceeds the sample size, the rank is capped at m, producing the historical maximum. This finite-sample reporting convention supplies no coverage guarantee under the observed dependence. Translated paths reanchor historical signed paths, while historical-peak quantiles are projected to the observed floor. The endpoint plug-in uses an absolute-displacement quantile; Table 3 gives the executed menus and schedules.

Ridge quantile regression links completed headroom labels to issue-known position states and calendar variables. Full regression uses 33 predictors plus an unpenalized intercept, whereas its four-feature variant uses magnitude, signed level and two root mean square (RMS) increment measures. Standardization uses only training rows with completed labels. Expanded-menu full regression changes only the penalty grid, using the following training-scaled objective:

$$(\hat{a}, \hat{b}) = \arg\min_{a,b} \left\{ \frac{1}{m} \sum_{i \in \mathcal{T}_r} \rho_\tau \left(\frac{Z_{i,H}}{s} - a - x_i^\top b \right) + \frac{\lambda}{2} \|b\|_2^2 \right\}, \qquad \lambda \in \{0.001, 0.01, 0.1, 1, 10, \infty\}.$$

The intercept is unpenalized; training rows alone determine the feature transform and response scale s. Reports are $B_t + s \max\{0, \hat{a} + x_t^\top \hat{b}\}$, while infinite shrinkage gives the ordinary empirical headroom quantile. The original menus and scale definitions remain in Table 3 and Supplement Sections S8 and S23.

Local quantiles use neighboring historical states, with an increment-scaled variant. Filtered paths simulate positions from GARCH-standardized empirical innovations drawn independently with replacement (Hull & White, 1998; Barone-Adesi et al., 1999). Quantile forests use leaf distributions; tail methods fit volatility-scaled headroom (Meinshausen, 2006; McNeil & Frey, 2000). Supplement Sections S7 and S11–S18 report benchmark results and failures; Section S34 links the complete legacy inventories.

Further comparisons apply delayed calibration, fixed combinations and conservative selection to the same peak response. A headroom CAViaR recursion and calibration wrappers follow existing methods (Engle & Manganelli, 2004; Gibbs & Candès, 2021). Calibration parameters update after outcomes mature. Fixed mean and median combinations use predefined members. Conservative selection favors the strongest shrinkage whose validation loss is within one paired HAC standard error of the winner's loss. This adapts one-standard-error selection (Hastie et al., 2009) to paired forecast losses.

Table 3: Forecast definitions and executed menus. Codes occur here to connect with the result files. Matched histories and update schedules make the reference explicit.

Method	Forecast definition	Executed menu / setting
Historical headroom (F)	Current floor plus a quantile of completed peak headrooms	Original FX histories {720,2880}; weekly {52,260}
Translated path (S)	Translate completed signed paths to the current position before taking their peaks	Same completed-label pools; fixed-window and rolling-window controls
Historical peak (K)	Historical absolute-peak quantile projected to the current floor	Same histories and update controls
Quantile regression (QR)	Training-scaled headroom regression with ridge slopes and an unpenalized intercept	Original three-point menu; six-point menu $\{0.001, 0.01, 0.1, 1, 10, \infty\}$
Local quantiles (KNN)	Neighbor headroom quantile, with a separate increment-scale variant	Training-only transforms and fixed neighbor menus; Supplement Sections S8 and S23
Four-feature regression (QR4)	Current signed/magnitude positions and short/long increment RMS	Original five-point menu; six-point paired comparison below
Endpoint plug-in (L)	Floor plus a high absolute-displacement endpoint quantile	Proposition 2; empirical plug-in evaluated separately
Further benchmarks	Autoregressive replay; filtered paths; peak-tail fit; quantile forest	Existing bounded specifications; results and complete failures linked from Supplement Sections S7 and S34
Coverage and combinations	Delayed projected ACI-style wrappers; headroom SAV; fixed forecast combinations	Original-source wrapper/SAV inventory and broader combination inventory are separate

Training histories count completed labels. Fixed-window models refit every 24 FX records or 13 weekly records. Quantile and normalization conventions are detailed in Supplement Sections S8 and S23; this table retains the executed menus.

6 Evaluation design

At refit time r, only labels satisfying $i + H \leq r$ are eligible; predictors use issue-known observations and training transforms remain fixed until the next refit. Original comparisons select parameters in the preceding validation segment, while the full-prefix backcast holds its specified penalties fixed. Original weekly comparisons use 52 outer validation origins with overlapping labels.

Matched comparisons hold the completed-response history and refit schedule fixed for regression and historical headroom. The main FX reference uses a shorter rolling history, whereas the main Treasury reference shares regression's longer history and refit schedule. Tables 4 and 5 report both alongside the other comparison sets.

Accuracy uses mean pinball loss and breaches are strict peak exceedances, scoring passive reports along the observed position paths. Paired loss differences use HAC covariance with the horizon-based bandwidth rule (Diebold & Mariano, 1995; Newey & West, 1987). Tables identify the small-sample adjustment and named multiplicity families (Harvey et al., 1997; Holm, 1979). Hit diagnostics use a fixed spaced grid and condition the transition likelihood on its initial hit (Kupiec, 1995; Christoffersen, 1998). For paired loss differences d_t, the comparison statistic and its existing bandwidth are

$$\mathrm{DM} = \frac{\sqrt{n}\,\bar{d}}{\sqrt{\widehat{\Omega}_K}}, \qquad K = \min\{n-1, \max[H, \lfloor 4(n/100)^{2/9} \rfloor]\},$$

where $\widehat{\Omega}_K$ is the Newey–West long-run variance estimate with Bartlett weights.

6.1 Research design and transparency

The original protocol fixes the two main comparisons and their reference procedures; the original FX evaluation period had already been inspected. Each new analysis states its prediction thresholds before computation (Supplement Section S36), with chronological records retained separately from the results. These later checks remain retrospective, and their descriptive fractions do not treat related markets as independent replications.

6.2 Multiple-method forecast comparison

We form a 90% model confidence set separately for each complete source, period, horizon and history grid (Hansen et al., 2011). The range statistic uses 1,000 stationary-bootstrap replications in arch 8.0.0, with block length $\max\{H, \lceil n^{1/3} \rceil\}$. All complete saved procedures enter on matching origin grids and targets, while incomplete prior methods remain explicitly unavailable. The nominal level presumes suitable loss dependence; it does not account for earlier research-stage model development.

Exact duplicate loss columns enter the bootstrap once, and their procedure labels are restored afterward. Historical reports are merged only after their full forecasts and targets match, while rolling references remain distinct. Several methods can produce identical losses, so one surviving loss pattern need not mean that only one method survives. The fractions summarize related tasks, with no independent cross-market pooled test.

7 Results

7.1 RQ1: matched accuracy and reference stability

Long-history FX regression beats matched historical headroom in both periods, yet its ranking against the shorter rolling reference reverses in the earlier period (Table 4). This distinction makes the reference procedure part of the substantive finding, since a conditioning gain alone does not determine the overall ranking.

Beyond the matched long-history case, short-history FX regression performs poorly at the longer horizon. Loss rises by 31.5% against matched historical headroom. Regression records 924 breaches against 427 for the reference; the full horizon map retains the other adverse comparisons. This adverse case shows why the matched long-history gain cannot be generalized across horizons and training-window lengths.

Delayed calibration retains the later matched FX gain, although its breach count remains slightly higher than wrapped historical headroom on common origins (Table 4). Calibration therefore changes breach frequency, while the distinct score and hit rankings persist on the common comparison grid.

Supplement Sections S20–S21 and S26 report the state, target-level and overlapping full-prefix comparisons (Tables S19–S21 and S25–S26).

Table 4: Long-history short-horizon FX regression under two references and periods. Negative percentages favor regression. Matched gains recur while the rolling-reference direction changes.

Period / procedure	Reference	Loss change (%)	Issues	Strict breaches (regression / reference)
Later / original main comparison	Shorter-history rolling headroom	-12.66	6625	376 / 332
Later / matched	Same-history fixed-window headroom	-14.77	6625	376 / 360
Earlier / original structure	Shorter-history rolling headroom	+3.17	1440	See complete earlier-period tables
Earlier / matched	Same-history fixed-window headroom	-16.62	1440	20 / 16
Later / coverage-wrapped	Same-history wrapped headroom	-15.00	6625	329 / 328

Regression uses history 2880 and horizon 6; the original rolling reference uses 720. The earlier matched comparison is the original three-penalty regression, not the later one-standard-error selection rule. The wrapper comparison is a separate finite projected delayed ACI-style exercise.

7.2 How much history does FX regression need?

The fixed-period, short-horizon learning curve shows that more history helps only over part of the tested range (Figure 2). Full regression moves from higher loss at the shortest history to its largest matched gain at an intermediate history. The gain is smaller at the longest history, while the four-feature curve changes sign within the tested range. Both curves use the original FX period and their original penalty menus; regression is compared with its own matched historical reference at each history length. Supplement Section S35 retains the full contrasts and synthetic confirmation summary. Together, the history and menu comparisons link conditional gains to history length and the availability of strong shrinkage.

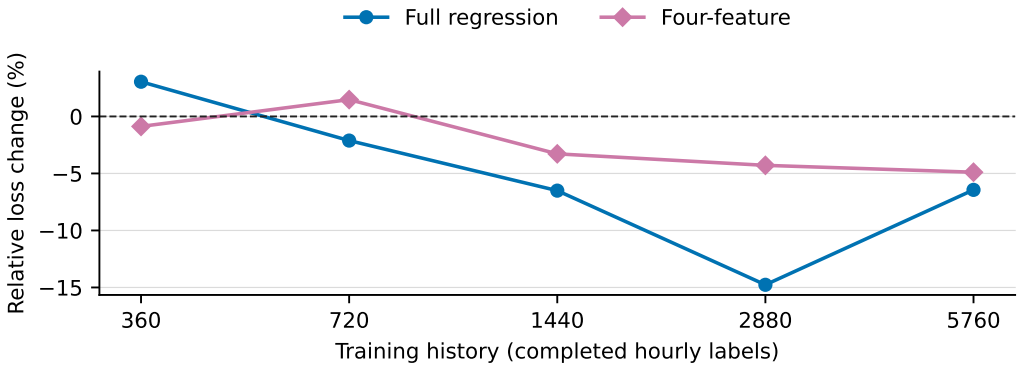


Figure 2: Saved short-horizon FX learning curves on the original evaluation period. The methods retain distinct original penalty menus, and every tested history is shown against its own matched historical headroom. This retrospective comparison does not identify a universally optimal history.

7.3 RQ2: daily, weekly and independently reported positions

The reference-sensitive hourly results motivate comparisons with other measurement clocks and reporting systems, each retaining matched histories and completed peak labels.

At weekly frequency, the Treasury-bill comparison favors historical headroom on both accuracy and breaches, extending the adverse ranking beyond the hourly examples. The earlier short-window version retains its separate adverse result because it uses a different history (Table 5).

Stronger regularization narrows weekly panel losses but leaves historical rules ahead, with conservative selection choosing the intercept-only report in 17 of 26 earlier cells (Table 5). Its largest relative loss increase occurs in the Treasury-bill short-history, 24-step comparison, despite conservative selection.

Neither simpler conditioning nor fixed combinations establish a uniform weekly improvement, reinforcing the need to retain the historical benchmark in each comparison. Supplement Sections S16–S18 retain their complete losses and failures, including the headroom-recursion results (Tables S14–S17).

Returning to the same FX record, historical headroom wins seven of eight daily comparisons; the exception uses four predictors, history 260 and horizon 6, reducing loss by 3.0%. It records 37 breaches versus 30 for matched headroom. Daily aggregation therefore preserves the score–breach disagreement while changing the leading forecast in most comparisons.

7.3.1 CFTC: independent reporting-system evidence

The independent reporting system gives a similar overall ranking: a historical headroom rule has lower loss in 29 of 34 contract/history comparisons. This meets the pre-specified 60% threshold, using the better realized score of corrected and ordinary empirical historical reports rather than an executable selector. Corrected historical headroom remains the fixed matched reference, with the ordinary empirical version reported separately. Both full-regression medians exceed the pre-specified ten-percent tolerance, whereas four-feature regression moves closer to history with the longer window. These comparisons extend the benchmark pattern to a separate reporting system while retaining substantial contract-level variation.

Table 5: Weekly evidence across the original main comparison, added panel and earlier-period comparisons. Counts refer to related tasks rather than independent dealers. Stronger regularization narrows losses while leaving substantial failures.

Comparison set	Procedure	Loss change / median (%)	Higher-loss cells
Original Treasury-bill main comparison	Original regression, long history	+50.79	Original main comparison
Earlier Treasury-bill short-window version	Short-history short-window version	+144.30	Different history
Additional panel	Original full regression	+92.63	45 / 48
Additional panel	Four-feature regression	+9.91	37 / 48
Additional panel	Full regression, expanded penalties	+9.22	34 / 48
Additional panel	Ordinary empirical headroom quantile	+0.50	30 / 48
Earlier weekly groups	Validation-minimum expanded rule	+4.15	26 groups
Earlier weekly groups	One-standard-error rule	+5.37	26 groups

The original bill main comparison has 54 versus 28 strict breaches among 277 issues. Earlier short-history bill results use a different history; the one-standard-error worst excess is +181.39%, and all its prewritten success criteria remain unmet. Supplement Sections S11–S15 list all panel and selection results (Tables S9–S13).

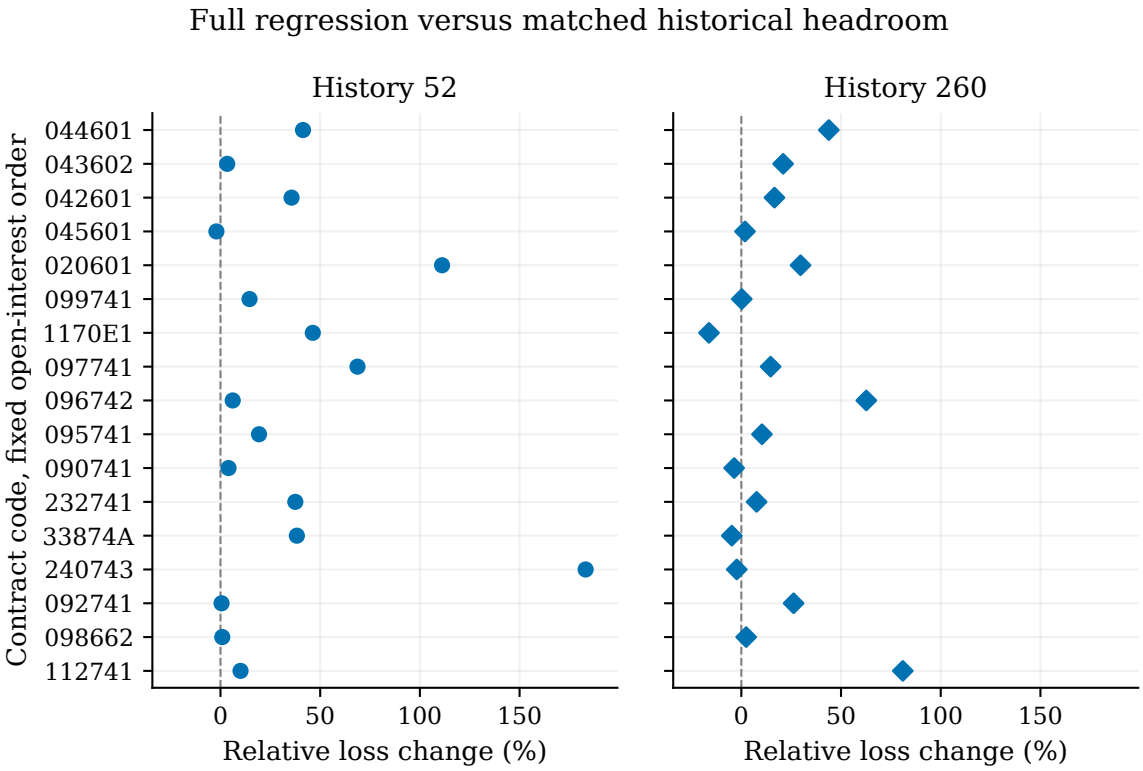


Figure 3: Expanded-menu regression relative loss versus matched historical headroom for every eligible CFTC contract. Columns separate histories; code order follows the fixed open-interest ranking. Positive values mean higher regression loss.

Table 6: CFTC dealer-category futures: relative loss versus matched historical headroom (%). All 17 selected literal contract codes are retained.

Contract	History 52					History 260				
code	Intercept only	Four-feature	Full regression	Calibrated history	Calibrated regression	Intercept only	Four-feature	Full regression	Calibrated history	Calibrated regression
044601	5.26	128.00	41.38	15.01	18.80	0.51	7.82	43.88	13.87	42.41
043602	3.37	9.35	3.37	37.79	24.64	2.30	8.95	20.97	-11.22	-12.64
042601	4.64	9.48	35.65	26.69	41.21	-0.15	74.23	16.63	1.46	9.39
045601	-2.04	0.31	-2.04	21.83	6.14	1.85	1.85	1.85	2.40	2.58
020601	4.77	79.72	111.12	25.66	-5.34	0.92	43.38	29.70	-15.12	-12.32
099741	0.69	13.97	14.56	77.33	-13.28	0.22	0.22	0.22	27.72	27.73
1170E1	6.42	6.82	46.30	-0.22	12.77	0.92	-28.44	-16.22	-1.12	1.00
097741	-0.83	23.76	68.71	29.77	33.79	0.53	-6.54	14.69	-0.41	-5.43
096742	2.93	3.41	6.13	49.54	30.46	5.30	4.82	62.68	37.12	-0.49
095741	7.39	19.55	19.33	41.27	42.98	-0.70	2.13	10.35	19.30	55.39
090741	4.03	10.28	4.03	15.58	13.13	3.28	-5.35	-3.60	-2.02	-5.66
232741	0.11	123.33	37.53	63.84	36.35	1.94	-3.35	7.68	-0.65	-3.86
33874A	6.95	88.57	38.30	30.15	44.07	-2.38	-3.21	-4.78	6.98	9.52
240743	-3.04	-7.32	183.12	33.14	49.77	-2.28	-2.28	-2.28	32.91	26.37
092741	0.54	0.54	0.54	-1.74	-0.68	0.17	-6.98	26.22	5.58	13.94
098662	-0.44	-0.44	0.90	44.89	38.90	2.45	-39.44	2.45	15.12	20.45
112741	11.06	38.76	10.05	33.72	24.10	2.24	2.82	80.98	8.36	16.37

Entries are percentage loss differences from matched corrected historical headroom; positive values mean higher loss, and unavailable entries retain failed or incomplete comparisons. The pre-specified conditions require historical superiority in at least 60% of all 34 comparisons, full-regression medians within ten percent at both histories, and a strictly smaller four-feature median with longer history. Full-precision results determine the comparisons, with all scores, breaches, selections and failures retained in Supplement Section S30.

7.3.2 Model confidence sets

Joint comparison leaves matched historical methods alongside alternatives in most sets, with a single historical method surviving in only a small minority. These results support retaining historical controls while leaving many alternatives uneliminated under the fixed bootstrap approximation. Non-elimination does not establish equal predictive risk. Table 7 reports surviving methods, with identical observed-loss patterns explained separately in Supplement Section S31. Supplement Section S31 retains every set, method p-value, equivalence class and prior unavailable method.

Table 7: Ninety-percent model confidence sets across all planned comparisons. Counts retain their comparison or conditional-procedure denominators.

Group	Complete / planned	Matched history included	Sole historical method	Conditional excluded	Unavailable procedures
FX original	4/4	2/4 (50.0%)	0/4 (0.0%)	39/66 (59.1%)	6
FX earlier period	4/4	3/4 (75.0%)	0/4 (0.0%)	1/15 (6.7%)	2
Treasury bills	4/4	4/4 (100.0%)	0/4 (0.0%)	11/74 (14.9%)	4
Other NYFed panels	48/48	46/48 (95.8%)	0/48 (0.0%)	110/384 (28.6%)	48
Same-record daily FX	4/4	4/4 (100.0%)	2/4 (50.0%)	5/8 (62.5%)	0
CFTC	34/34	31/34 (91.2%)	0/34 (0.0%)	31/136 (22.8%)	0
All comparisons	98/98	90/98 (91.8%)	2/98 (2.0%)	197/683 (28.8%)	60

Matched history includes corrected and ordinary empirical headroom reports, excluding rolling references and nearest-neighbor baselines; conditional exclusions count complete procedures. Identical losses can leave several methods tied, so the sole-method column counts only sets containing exactly one historical procedure. Supplement Section S31 retains all tied loss patterns, secondary membership counts and unavailable procedures; related comparisons are not independent replications.

7.4 Why does regression lose at low frequency?

Short weekly samples leave both tuning errors and a performance gap within the finite penalty menu. Figure 4 splits full regression's actual excess into the selected report's gap from the best finite candidate and that candidate's gap from historical headroom. The best candidate uses evaluation hindsight over five finite penalties; the ordinary empirical option is excluded. For full regression, the finite-menu gap is positive in 29 of 41 short-history weekly cells; selection exceeds half of positive actual excess in 27. Both criteria hold, so benchmark advantages and selection losses coexist within this menu. Separate component medians need not add: actual excess is 19.44%, selection 18.32%, and the diagnostic gap 4.37%.

This retrospective finite oracle is diagnostic, cannot guide deployment, and cannot rule out information in other conditional models. When validation selects the ordinary empirical limit, it can beat every finite candidate and make the selection component negative. Both long-history FX horizons have a negative finite-menu gap. Expanded menus also bring the calibrated simulation medians close to their historical reference, leaving the empirical frequency mechanism unidentified.

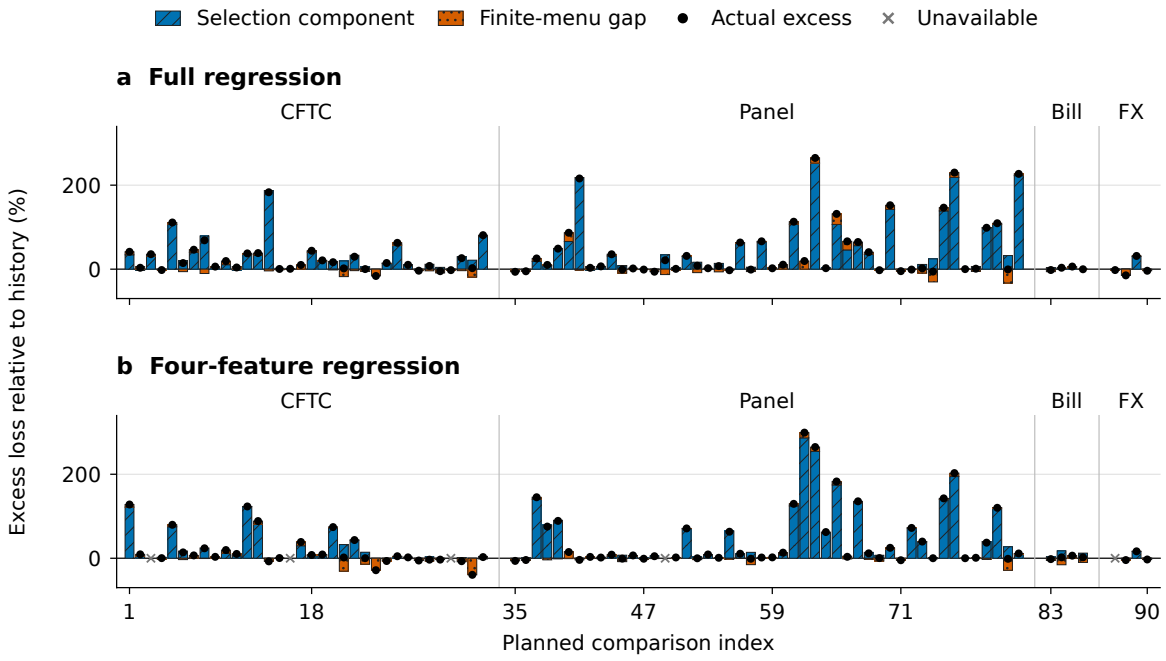


Figure 4: Signed decomposition of regression excess loss against matched historical headroom, with selection in blue, the finite-menu diagnostic gap in orange, and actual excess as black dots. Crosses retain unavailable comparisons; the text focuses on short-history weekly cells while all planned comparisons remain plotted.

Extending validation from 52 to 156 weeks keeps the CFTC evaluation period, training histories, and refitting clock fixed. Median excess falls from 19.33% to 4.03% with 52-week training histories, and from 10.35% to 0.53% with 260-week histories. Infinite-penalty selections rise from 4 to 10 of 17 contracts, and from 4 to 5 of 17, respectively. These differences between across-contract medians need not match typical paired improvements and do not identify a causal effect. Four-feature failures remain unavailable in the complete inventory (Supplement Section S36).

Tail-level sensitivity remains incomplete, as Table 8 shows for 90% and 99% quantiles under matched histories. At 99%, historical headroom wins 28 of 34 CFTC comparisons; one failure leaves the 29-win threshold unresolved. Both FX periods remain unavailable at 99%, leaving the

combined improvement criterion undetermined. Unavailable planned comparisons are retained, and successful subsets never replace a missing full-group median.

Table 8: Full-regression relative excess loss (%) at alternative quantile levels. Positive values favor matched historical headroom; NA retains any failed required comparison.

Source / training history	90%	99%
CFTC, 52 weeks	+3.45	+15.05
CFTC, 260 weeks	+1.58	NA
FX, original period	-10.15	NA
FX, earlier period	-17.85	NA
Treasury bills	+0.67	+44.22

Contract rows report full-group medians. FX uses 2,880-hour training histories; Treasury uses 260 weeks. Successful subsets never replace a required full group.

7.5 Scores and breach behavior answer different questions

Accuracy rankings leave breach behavior unresolved: the matched FX gain coexists with more breaches before calibration (Table 4). Pinball loss measures error amounts while a strict-hit indicator records event occurrence, so monitoring reports should show both forms of error.

Elementary-score rankings cross for both original main comparisons (Supplement Figures S2–S3, Section S33). A separate 24-step FX comparison against rolling translated paths reverses near 20.6:1. Both six-step FX comparisons reduce mean underprediction and mean overprediction, so the weight reversal belongs specifically to the 24-step reference.

Loss and hit diagnostics answer different questions (Supplement Table S47, Section S33). HAC tests compare average loss and strict-hit tests examine event frequency and serial behavior, while RMS-state diagnostics compare conditional predictive ability across training-defined groups (Giacomini & White, 2006).

8 RQ3: limit-cost regions

$$J_m(\kappa,\beta,c)=\kappa\frac{\overline{C_m-B}}{m_F}+\beta\,\overline{\mathbf{1}\{Y>C_m\}}+c\frac{\overline{(Y-C_m)_+}}{m_F},\qquad m_F=\overline{C_F-B}>0,\quad \beta=b/m_F.$$

For CFTC, higher reserve coefficients and lower fixed-breach coefficients generally favor expanded-menu regression over matched historical headroom, although the grid is not monotone (Figure 5). At $c=1$, regression is strictly cheaper in 5/34 comparisons at the center of the sorted grid $(\kappa,\beta)=(.1,.25)$, 32/34 at $(1,0)$, and 1/34 at the opposite corner $(.01,5)$. These are illustrative costs, not observed operating costs; Supplement Section S32 gives the normalization, fixed grid and complete-method results.

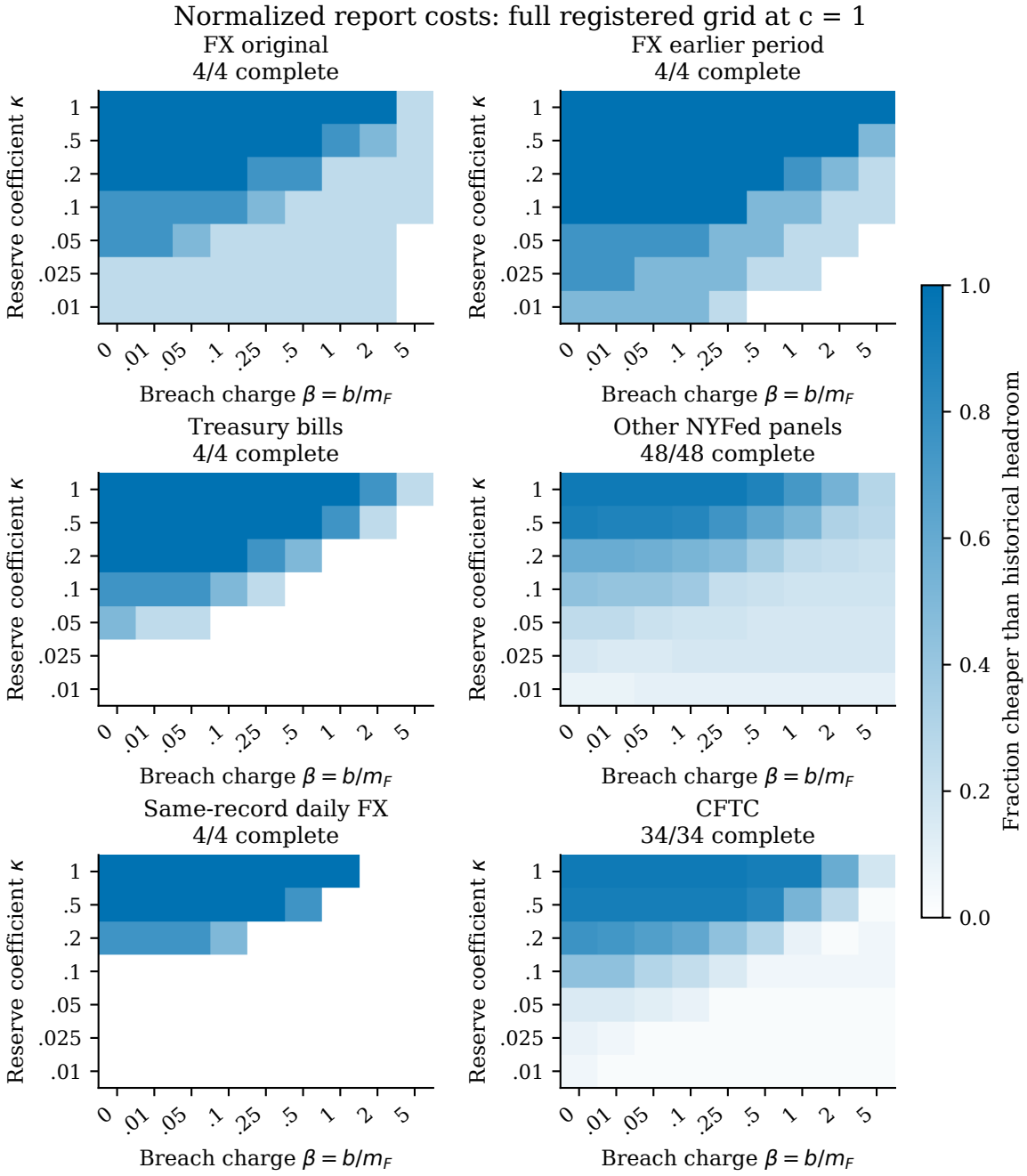


Figure 5: Fixed reserve and breach-charge regions with the exceedance coefficient held at unity. Color shows the fraction of related comparisons where the stated conditional procedure costs less than corrected historical headroom, with each source group retaining its own histories and horizons. These are descriptive fractions; all comparisons are complete, and the lower and upper fractions coincide.

9 Implications for model validation

Benchmark against historical headroom. Use the same completed labels, history length and refit schedule, naming the shorter rolling reference separately because its FX ranking changes across periods. This keeps differences in available history distinct from the gains associated with conditioning within a matched comparison.

Use historical headroom by default when validation evidence is short. Expanded penalties and longer validation reduce losses, but short-sample selection, including one-standard-error selection, still fails to protect the historical benchmark consistently. Historical reports also survive in most CFTC model confidence sets, supporting their retention until a conditional procedure has sufficient validation evidence. The ordinary empirical quantile is up to 36.5% worse than corrected headroom in the earlier weekly grid.

Report breaches with the score. Show mean underprediction, mean overprediction and strict-breach counts together, since the unwrapped matched FX gain and daily regression exception both accompany more breaches than history. Cost regions add reserve, breach-frequency and exceedance preferences to this joint evaluation.

10 Limitations

- FX comes from one market maker, and its original evaluation period was inspected before conditional methods were specified.
- The earlier-period comparison and full-prefix backcast are retrospective, with no previously unseen prospective financial evaluation.
- Weekly reports aggregate positions across dealers; related categories and the CFTC survivor universe do not provide independent dealer replications.
- Downloaded economic-date vintages omit historical first-release information, reporting delays and unrecorded within-interval excursions.
- The comparisons identify neither causal effects nor individual-feature effects, and fitted limits receive no coverage guarantee under this dependence.
- Costs illustrate fixed preferences; trading returns, funding gains and institutional intervention costs are not measured.
- The finite-penalty hindsight bound is diagnostic only; validation-length and quantile-sensitivity checks are post-result analyses with numerical failures retained.

Supplement Section S36 retains the original technical qualifications on simulations, aggregation and forecast comparison.

11 Conclusion

Historical rules are hard to beat. For RQ1, matched FX regression gains recur across periods, but the shorter rolling reference changes the ranking and history gains are not monotone. Delayed calibration preserves the later matched gain with a small breach-count difference, reinforcing the need to interpret conditioning gains under explicit histories and update schedules.

For RQ2, historical controls remain competitive across related weekly aggregates and same-record daily FX, usually surviving joint comparison alongside alternatives. The finite-penalty hindsight diagnostic leaves losses in most short weekly comparisons, while longer validation improves CFTC selection toward historical reports. Both menu limits and selection errors therefore matter within the tested procedures; neither result rules out every possible conditional signal.

Expanded simulation menus bring short-history regression near its historical reference, with switching adding no further observed median loss in that comparison. Numerical failures prevent a complete assessment at the higher quantile level, leaving the original gains without a demonstrated extension there.

For RQ3, higher reserve charges and lower fixed breach charges generally favor expanded regression in CFTC, with local exceptions. Lower forecast scores therefore need not imply lower limit costs. Validation should report reserve amounts, breach counts and exceedance amounts together, using the cost regions in Figure 5.

Data and code availability

Data availability. The FX input is the specified version of the Kaggle Net Position Time Series Dataset. The owning metadata labels its license "Data files © Original Authors." This repository supplies the acquisition route and input fingerprint, but excludes raw FX observations and private per-issue prediction vectors. Obtain the source from `https://www.kaggle.com/datasets/davidlindsay1979/net-position-time-series-dataset` subject to the original rights holders' terms.

The New York Fed aggregates are available from its Primary Dealer Statistics service under the Bank's Terms of Use. The repository preserves the unchanged accepted Treasury snapshot with attribution and identifies study calculations separately. Copying or distributing that content remains subject to its source notices, attribution, modification labeling, continuing conditions and endorsement restrictions: `https://www.newyorkfed.org/privacy/termsofuse.html`. The code license does not replace those source conditions.

The CFTC application uses the public TFF historical files from `https://www.cftc.gov/MarketReports/CommitmentsofTraders/HistoricalCompressed/index.htm`. The repository includes unchanged downloaded archives and a per-file URL, acquisition time, byte count and SHA-256 record in `data/cftc-tff/provenance.json`.

Code availability. Code, results and reproduction instructions are available at `https://github.com/ByronLi0207/financial-mathematics-paper-20261020/tree/revision-v12`.

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Author information: [TO BE COMPLETED BY AUTHOR].

ORCID: [TO BE COMPLETED BY AUTHOR].

Funding: [TO BE COMPLETED BY AUTHOR].

Competing interests: [TO BE COMPLETED BY AUTHOR].

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